AR Parameter estimation

$$y[t] = -a_1 y[t-1] - a_2 y[t-2] - \dots - a_p y[t-p] + f[t]$$

when $f[t]$ is
- zero-mean
- white
- variance σ_f^2

AR(p)

autoregressive

1. Mult.
2. Take expectations

$$E(y[t]\, y[t-\ell]) = E[-a_1 y[t-1] y[t-\ell] - a_2 y[t-2] y[t-\ell] - \dots - a_p y[t-p] y[t-\ell] + f[t]\, y[t-\ell]]$$

$$r_y[\ell] = -a_1 r_y[\ell-1] - a_2 r_y[\ell-2] - \dots - a_p r_y[\ell-p] + E[f[t]\, y[t-\ell]]$$

Expectations

Ex: X: throw of a die

$P(X=1) = \frac{1}{6}$

$P(X=2) = \frac{1}{6}$

$\vdots$

$P(X=6) = \frac{1}{6}$

$$E[X] = 1 \cdot P(X=1) + 2\, P(X=2) + \dots + 6\, P(X=6)$$

$$= \frac{1}{6}(1+2+3+4+5+6)$$

$$= \underline{3.5}$$

Ex: X = swap takes

$P(X = 10^6)$

$P(X = 0)$

$$E[X] = 10^6 \cdot \underbrace{P(X=10^6)}_{10^{-7}} + 0 \cdot \underbrace{P(X=0)}_{1-10^{-7}}$$

$$= 10^{-1}$$

Discrete $\quad E[X] = \sum_x x\, P(X=x)$

Cont: $\quad E[X] = \int_{-\infty}^{\infty} x\, p_X(x)\, dx$

- Expectations are linear operators.

$$E[a_1 X_1 + a_2 X_2 + \dots + a_n X_n] = E[a_1 X_1] + E[a_2 X_2] + \dots + E[a_n X_n]$$

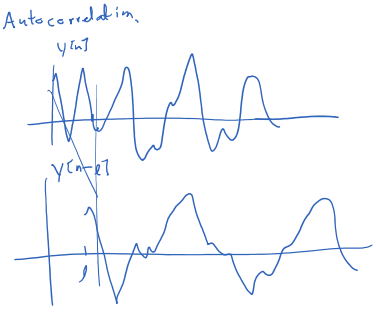
- Non random factors can be pulled out of expectations.

$$= a_1 E[X_1] + a_2 E[X_2] + \dots + a_n E[X_n]$$

$y[n]$: random process.

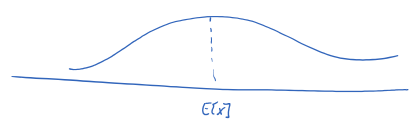
$E[y[n]]$ = mean.

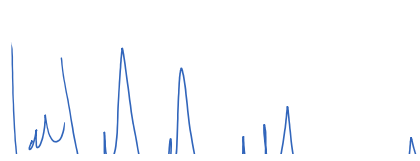
Autocorrelation.

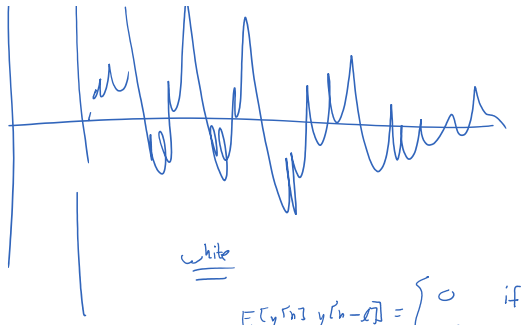


Mean: $E[X]$

Variance: $\sigma^2 = E[(X - E[X])^2] = E[X^2] - (E[X])^2$







$r_y(\ell) = r(\ell) = E[\, y[n]\, y[n-\ell] \,]$

$r_{yy}(\ell)$

white

$$E[y[n]\, y[n-\ell]] = \begin{cases} 0 & \text{if } \ell \neq 0 \\ \sigma_y^2 & \text{if } \ell = 0 \end{cases}$$

$$\boxed{r_y[\ell] = -a_1\, r_y[\ell-1] - \cdots - a_p\, r_y[\ell-p]} + \underbrace{E[\, y[t-\ell]\, f[t] \,]}_{\cancel{0} \text{ if } \ell > 0}$$

$$\begin{array}{c} \ell=1 \\ \ell=2 \\ \\ \ell=p \end{array} \underbrace{\begin{bmatrix} r_y[0] & r_y[-1] & \cdots & r_y[1-p] \\ r_y[1] & r_y[0] & \cdots & r_y[2-p] \\ \vdots & & & \\ r_y[p-1] & r_y[p-2] & \cdots & r_y[0] \end{bmatrix}}_{R} \underbrace{\begin{bmatrix} -a_1 \\ -a_2 \\ \vdots \\ -a_p \end{bmatrix}}_{(\,q\,)} = \underbrace{\begin{bmatrix} r_y[1] \\ r_y[2] \\ \vdots \\ r_y[p] \end{bmatrix}}_{p}$$

Toeplitz matrix

Yule-Walker

Real: $R = R^T$ (symmetric)

Complex: $R = R^H$

H: conjugate transpose.

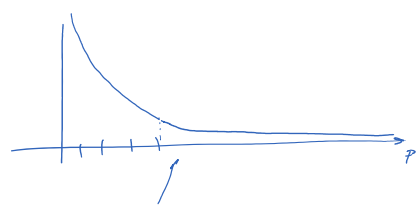
$R\, q = p. \qquad q = R^{-1} p.$

$[\, r[0] \,]$

$\begin{bmatrix} r[0] & r[-1] \\ r[1] & r[0] \end{bmatrix}$

$\begin{bmatrix} r[0] & r[-1] & r[-2] \\ r[1] & r[0] & r[-1] \\ r[2] & r[1] & r[0] \end{bmatrix}$

$E[(y[n] - \hat{y}_p[n])^2]$



$(n+1) \times (n+1)$:

$$T_n = \begin{bmatrix} r_0 & r_1 & \cdots & r_n \\ r_1 & r_0 & \cdots & r_{n-1} \\ \vdots & & & \\ r_n & & \cdots & r_0 \end{bmatrix} \qquad T_0 = r_0 \qquad T_1 = \begin{bmatrix} r_0 & r_1 \\ r_1 & r_0 \end{bmatrix}, \text{ etc.}$$

$$T_n \underbrace{\begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_{n+1} \end{bmatrix}}_{\underline{w}_n} = \underbrace{\begin{bmatrix} r_1 \\ r_2 \\ \vdots \\ r_{n+1} \end{bmatrix}}_{\underline{r}_n}$$

Fact: Let $J = \begin{bmatrix} 0 & & & 1 \\ & & 1 & \\ & 1 & & \\ 1 & & & 0 \end{bmatrix}$

If T is a Toeplitz matrix then $T = J\, T^T J$

$[x_1] \quad [x_n]$

If T is a <u>Toeplitz</u> matrix then $\underline{\underline{T = \underbrace{J T^T J}_{\text{Persymmetric}}}}$

pf: $J \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} x_n \\ x_{n-1} \\ \vdots \\ x_1 \end{bmatrix}$

$[x_1 \cdots x_n] J = [x_n \; x_{n-1} \cdots x_1]$

$$J T^T = J \begin{bmatrix} r_0 & r_1 & \cdots & r_n \\ \vdots & & & \\ r_{n-1} & r_{n-2} & & \\ r_n & r_{n-1} & & r_0 \end{bmatrix} = \begin{bmatrix} r_n & r_{n-1} & & r_0 \\ r_{n-1} & r_{n-2} & \cdots & \vdots \\ \vdots & \vdots & & \\ r_0 & r_1 & & r_n \end{bmatrix}$$

$$J T^T J = \begin{bmatrix} r_0 & r_1 & \cdots & r_n \\ r_1 & r_0 & \cdots & r_{n-1} \\ \vdots & & & \\ r_n & \cdots & & r_0 \end{bmatrix}$$

$$T = J T^T J$$

$$\begin{cases} J T = J^2 T^T J \\ \quad\;\; = T^T J \end{cases}$$

$$(JT)^{-1} = (T^T J)^{-1}$$

$$= T^{-1} J^{-1} = J^{-1} T^{-T}$$

$$= T^{-1} J = J T^{-T}$$

$$T J = J T^T J^2 = J T^T$$

$$(TJ)^{-1} = (J T^T)^{-1}$$

$$\underline{J T^{-1} = T^{-T} J}$$

Inductively, suppose we have the solution $T_n \underline{w}_n = \underline{r}_n$

want to do: use this to find the solution to $T_{n+1} \underline{w}_{n+1} = \underline{r}_{n+1}$

$$\underline{w}_{n+1} = \begin{bmatrix} \underline{z}_n \\ \alpha_n \end{bmatrix}$$

$$T_{n+1} = \left[\begin{array}{c|c} T_n & \begin{matrix} r_{n+1} \\ r_n \\ \vdots \\ r_1 \end{matrix} \\ \hline r_{n+1} \; r_n \cdots r_1 & r_0 \end{array}\right] = \left[\begin{array}{c|c} T_n & J_n \underline{r}_n \\ \hline \underline{r}_n^T J_n & r_0 \end{array}\right]$$

$$T_{n+1} \underline{w}_{n+1} = \underline{r}_{n+1}$$

$$\left[\begin{array}{c|c} T_n & J_n \underline{r}_n \\ \hline \underline{r}_n^T J_n & r_0 \end{array}\right] \begin{bmatrix} \underline{z}_n \\ \alpha_n \end{bmatrix} = \begin{bmatrix} \underline{r}_n \\ r_{n+2} \end{bmatrix}$$

$$\begin{cases} T_n \underline{z}_n + J_n \underline{r}_n \alpha_n = \underline{r}_n \\ \underline{r}_n^T J_n \underline{z}_n + r_0 \alpha_n = r_{n+2} \end{cases} \longrightarrow \underline{z}_n = T_n^{-1} (\underline{r}_n - J_n \underline{r}_n \alpha_n)$$

$$= \underline{w}_n - T_n^{-1} J_n \underline{r}_n \alpha_n$$

$$= \underline{w}_n - J_n T_n^{-T} \underline{r}_n \alpha_n$$

$$= \underline{w}_n - J_n \underline{w}_n \alpha_n$$

$$\underline{r}_n^T J_n (\underline{w}_n - J_n \underline{w}_n \alpha_n) + r_0 \alpha_n = r_{n+2}$$

$$(-\underline{r}_n^T \underline{w}_n + r_0) \alpha_n = r_{n+2} - \underline{r}_n^T J_n \underline{w}_n$$

$$\boxed{\alpha_n = \frac{r_{n+2} - \underline{r}_n^T J_n \underline{w}_n}{}}$$

$$\boxed{\alpha_n = \frac{r_{n+2} - \underline{r}_n^T J_n \underline{w}_n}{r_0 - \underline{r}_n^T \underline{w}_n}}$$

In Matlab, ' is conjugate transpose.

.' is transpose.

$$E\Big[\underbrace{\big(y[n] + a_1 y[n-1] + \dots + a_p y[n-p] - f[n]\big)}_{\text{error}} \underbrace{y[n-\ell]}_{\text{"data"}}\Big] = 0$$

$$\underline{y}[n-1] = \begin{bmatrix} y[n-1] \\ y[n-2] \\ \vdots \\ y[n-p] \end{bmatrix} \qquad \underline{a} = \begin{bmatrix} -a_1 \\ -a_2 \\ \vdots \\ -a_p \end{bmatrix}$$

$$\hat{y}[n] = -a_1^* y[n-1] - \dots - a_p^* y[n-p]$$
$$= \underline{a}^H \underline{y}[n-1]$$

choose $\{a_1, a_2, \dots, a_p\}$ so that $E\left[\left(y[n] - \hat{y}[n]\right)^2\right]$ is small.

$$\downarrow$$

$$\min_{\{a_1, \dots, a_p\}} E\left[\left(y[n] - \underline{a}^T \underline{y}[n-1]\right)^2\right]$$

$$E\left[y[n]^2 - 2\, y[n]\, \underline{a}^T \underline{y}[n-1] + \underline{a}^T \underline{y}[n-1]\, \underline{y}^T[n-1]\, \underline{a}\right]$$

$$= E[y[n]^2] - 2\underline{a}^T \underbrace{E[y[n]\underline{y}[n-1]]}_{\underline{p}} + \underline{a}^T \underbrace{E[\underline{y}[n-1]\, \underline{y}^T[n-1]]}_{R}\, \underline{a}$$

$$= E[y[n]^2] - 2\underline{a}^T \underline{p} + \underline{a}^T R\, \underline{a}$$